

The Bridge Study — Supplement (v3.1)

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Companion to the **Version 3.1** report — August 18, 2026

Companion to `docs/research/BRIDGE-REPORT.md` (version 3, 2026-08-02; typeset as `report.pdf`). Everything the main paper’s structure moved out lives here: the complete preregistration execution inventory, raw-instrument tables and outage bases, instrument-validation detail, exposure strata, sensitivity analyses, the full tool-call census, the WF manual, review responses, the changelog, and data availability. Section numbers §Sn are cited from the main paper. Nothing here is new evidence; every table is recomputable from the file named beside it.

S1 Complete preregistration execution inventory

Table 1 lists every preregistered component, its status, and the inferential consequence. Its sources are the preregistration (`docs/research/BRIDGE-EXPERIMENT.md`, commit `0d36f266`) and the claim-by-claim inventory in `docs/research/review/verification-of-review-1.json` (prereg section).

Table 1: Preregistration execution inventory.

Preregistered component	Status	What actually happened	Inferential consequence
Five arms A–E, identical model/prompt/budget, tools-only difference	executed	471 rows/arm (<code>bench/data/runs/</code>)	design intact
Per-tool call logging	executed	<code>tool_calls_by_name</code> every row; full traces eval C/D/E + fresh (deviation 7)	enables trace attribution
Tier 1a ProofNet# (371)	executed	371/arm, scored	contaminated by construction (arm A 59.8% on eval-341); paired deltas only
Tier 1b fresh set (~100)	executed	100 tasks	holdout narrower than claimed (Brain indexes only; one later-fold leak) — renamed post-Brain-index, 99/100
Hallucinated-decl rate	executed	all tables	instrument required repair (§S3); repaired rates are the report’s run on grounded typecheck, not the preregistered endpoint
McNemar D-vs-E / D-vs-C + discordant tables	executed	main §4	—
10-min wall clock	executed	600 s timeouts	—
Pre-campaign API requirements 1–8	executed	implemented before campaign	—
30-turn budget	modified	advisory only; overruns C 50/D 38/E 48 on the repaired rows (E 32 as-run — outage rows masked overruns), max 88	budget uncontrolled; §S5 sensitivity
Determinacy pre-screen (exclude)	modified	post-hoc dual annotation (79%, $\kappa \approx 0.20$); 74-task subset, none excluded	subset analysis, not screened population
Arm-D production server	modified	production Worker code served locally over shipped shards	none material
Single-shot elaboration grading	modified	persistent REPL, same pins	none material
≤4 typecheck calls in-loop	modified	no in-loop typechecking at all	halluc rate is a pure grounding measure
Skill/ToolSearch leak	modified	sealed before eval; 120 dev runs quarantined	dev split discarded, eval clean
Per-arm cost distributions	modified	means only (USD + wall-clock)	descriptive (§S10)

Preregistered component	Status	What actually happened	Inferential consequence
faithful@budget (BEq+ equivalence)	not executed	grader a stub; zero judge files in campaign	primary endpoint missing; zero confirmatory results
LLM judge + 50-item human calibration	not executed (campaign)	harness existed unused; an <i>uncalibrated</i> blind judge pass ran post-review (§S11)	main §4.2 is exploratory only
tokens-to-solve (half the success criterion)	not executed	never computed (tokens recorded per row)	P2 untested as specified
3 reseeds + pass@k curves	not executed	one run per (task, arm) everywhere	no variance estimate anywhere
Second model class on primary set	not executed	Tier-1 all-Haiku; v2 all-Sonnet on different benchmarks	capability-band generality unknown
Preregistered success criterion	not executed	neither half graded as specified	zero confirmatory results
Tier 2 as specified (FATE-H + MPR-Prop proving, reflection loop)	not executed	replaced by exploratory v2: QR/MPR retrieval + one-shot SorryDB, arms N/F/W/U/WF	main §§5–6 exploratory
PutnamBench	not executed	—	—
Tier 3a offline Erdős set	not executed	—	—
Tier 3b live Erdős queue	not executed	—	—

S2 Tier-1: raw-instrument tables and analysis bases

Sources for this section, all under `bench/analysis/`, are `tier1_reanalysis.{py,json,md}`, `part1_fresh100_v2.{json,md}`, `fresh_clustered.{py,json,md}`, and `success_repaired.{py,json,md}`. Grading pins: eval-341 on Lean v4.32.0-rc1 / Mathlib a33a5ccd; fresh-100 on Lean v4.33.0-rc1 / Mathlib 9944fe29; the checkout C/E’s file tools read is 61a5e4f338 (content ~2026-07-10); ProofNet# source pin PAug/ProofNetSharp @ a8da405f (MIT).

S2.1 The three arm-E bases (raw instrument)

E’s 31 infrastructure-dead rows (fresh_069–099, contiguous session-limit 429s; A–D verified zero errors) can be counted three ways. Cells show grounded typecheck as $n/N = \%$ [Wilson 95% CI], on the **raw oracle**:

arm	errors-as-failures (n=100)	completed pairs (n=69)	post-repair (n=100)
A	20/100 = 20.0% [13.3, 28.9]	10/69 = 14.5% [8.1, 24.7]	20/100 = 20.0% [13.3, 28.9]
B	22/100 = 22.0% [15.0, 31.1]	12/69 = 17.4% [10.2, 28.0]	22/100 = 22.0% [15.0, 31.1]
C	25/100 = 25.0% [17.5, 34.3]	12/69 = 17.4% [10.2, 28.0]	25/100 = 25.0% [17.5, 34.3]
D	42/100 = 42.0% [32.8, 51.8]	28/69 = 40.6% [29.8, 52.4]	42/100 = 42.0% [32.8, 51.8]
E	16/100 = 16.0% [10.1, 24.4] — 31 infra errors	16/69 = 23.2% [14.8, 34.4]	30/100 = 30.0% [21.9, 39.6]

Raw McNemar (exact binomial two-sided): errors-as-failures D-vs-E 32/6 $p=2.4e-5$ (conflates outage with behavior — quarantined here); completed-69 D-vs-E 18/6 $p=0.023$, D-vs-C 21/5 $p=0.0025$, D-vs-A 22/4 $p=0.0005$; post-repair full-100 D-vs-E 25/13 $p=0.073$, D-vs-C 28/11 $p=0.0095$, D-vs-A 29/7 $p=0.0003$, E-vs-A 19/9 $p=0.087$. Of the 31 repaired E rows, 14 became successes.

Wald/McNemar mismatch note. On the post-repair full-100 raw table, the paired-Wald D–E risk-difference interval was $[+0.0015, +0.2385]$ (barely excluding 0) while exact McNemar gave $p=0.073$ — a normal approximation over all 100 paired differences versus a test conditioned on the 38 discordant pairs, landing on opposite sides of $\alpha=.05$ near the boundary. This is why v3 replaces both with a single commit-clustered bootstrap whose interval and p-value are read off the same resampling distribution.

S2.2 Raw vs repaired instrument, side by side

Repaired oracle = `halluc_validation.py classify_adjusted` (rules R1–R5, §S3), folded into success by `success_repaired.py`; success is monotone under the repair (flags are only ever cleared). Affected rows (raw-flagged, repaired-clean) and how many then typecheck:

arm	affected (fresh)	typecheck pass	affected (eval-341)
A	17	1	39
B	18	0	45
C	38	8	68
D	17	6	47
E	39	7	81
total	129	22	280

arm	raw k/n [Wilson 95]	repaired k/n [Wilson 95]	Δ
A	20/100 [13.3, 28.9]	21/100 [14.2, 30.0]	+1
B	22/100 [15.0, 31.1]	22/100 [15.0, 31.1]	+0
C	25/100 [17.5, 34.3]	33/100 [24.6, 42.7]	+8
D	42/100 [32.8, 51.8]	48/100 [38.5, 57.7]	+6
E	30/100 [21.9, 39.6]	37/100 [28.2, 46.8]	+7

McNemar, both instruments:

pair	raw b/c	raw p	repaired b/c	repaired p	classification
D vs E	25/13	0.073	28/17	0.135	ns $\rightarrow$ ns
D vs C	28/11	0.0095	32/17	0.044	sig $\rightarrow$ sig
D vs A	29/7	0.00031	34/7	2.5e-5	sig $\rightarrow$ sig
C vs A	16/11	0.442	23/11	0.058	ns $\rightarrow$ ns
E vs A	19/9	0.087	24/8	0.007	ns $\rightarrow$ sig

Commit-clustered paired bootstrap (44 clusters, B=10,000; raw column asserted byte-identical to `fresh_clustered.json` before the repaired column was computed):

pair	raw RD [95% CI] p	repaired RD [95% CI] p	classification
D - E	+0.120 [-0.049, +0.271] p=0.192	+0.110 [-0.089, +0.279] p=0.304	ns $\rightarrow$ ns
D - C	+0.170 [+0.011, +0.314] p=0.040	+0.150 [-0.023, +0.302] p=0.100	sig $\rightarrow$ ns
D - A	+0.220 [+0.093, +0.336] p=0.0014	+0.270 [+0.131, +0.395] p=0.0004	sig $\rightarrow$ sig

Note the two classification changes at $\alpha=.05$ cut in opposite directions (E-vs-A strengthens, D-vs-C weakens) — the repair is not a thumb on either scale.

S2.3 Cluster census and sensitivity collapses

The 100 tasks decompose into **44 distinct source commits**, **57 files**, and **59 name-stem families** (24 multi-member). The largest commit clusters are **87a6eccf** (9 tasks, the AntitoneOn sum-integral family), **49ed1b2d** (8, bounded variation), **61303857** (5), two of size 4. On the raw instrument, +7 of D’s net +12 paired advantage over E comes from the 9-task commit (D 7/9, E 0/9) and +3 from the 8-task commit (D 4/8, E 1/8) — 83% of the net effect in 2 of 44 commits.

All pairs $\times$ all clustering levels (raw instrument, paired bootstrap RD [95% CI] p):

pair	task-level (iid)	family-clustered	file-clustered	commit-clustered
D - E	+0.120 [+0.000, +0.240] p=0.062	+0.120 [-0.044, +0.274] p=0.186	+0.120 [-0.044, +0.274] p=0.178	+0.120 [-0.049, +0.271] p=0.192
D - C	+0.170 [+0.050, +0.290] p=0.0064	+0.170 [+0.021, +0.311] p=0.032	+0.170 [+0.021, +0.310] p=0.030	+0.170 [+0.011, +0.314] p=0.040
D - A	+0.220 [+0.110, +0.330] p=0.0002	+0.220 [+0.080, +0.346] p=0.0028	+0.220 [+0.080, +0.344] p=0.0026	+0.220 [+0.093, +0.336] p=0.0014

Family/file/commit-collapsed sensitivity (one unit per cluster, majority and any-success rules) is tabulated in `bench/analysis/fresh_clustered.md` §4; under every collapse D–E is a null ($p \geq 0.52$), D–A survives the any-success collapses ($p=0.027$ – 0.041), and the tie-free mean collapse gives D–E +0.043 [–0.104, +0.183] $p=0.554$, D–A +0.157 [+0.043, +0.264] $p=0.0064$.

S2.4 Eval-341 (ProofNet#) raw table and the repaired-instrument gap

arm	grounded typecheck (raw)	Wilson 95% CI
A	204/341 = 59.8%	[54.5, 64.9]
B	198/341 = 58.1%	[52.8, 63.2]
C	218/341 = 63.9%	[58.7, 68.9]
D	219/341 = 64.2%	[59.0, 69.1]
E	208/341 = 61.0%	[55.7, 66.0]

The 280 raw-flagged repaired-clean eval rows (per-arm counts in S2.2) exceeded the typecheck budget and were **not** re-typechecked; the counts are hard upper bounds on per-arm gains, and the instrument-bias direction (C/E over-flagged relative to D) applies to this table too. Typecheck alone anti-correlates with grounding on this split (E led typecheck at 83.3% while trailing on hallucinations), reproducing TheoremGraph’s typecheck-is-not-a-signal finding at $15\times$ their n.

S3 Hallucination-oracle validation detail

Sources for this section are `bench/analysis/halluc_validation.{py,json,md}`, with blinded intermediates in `bench/analysis/halluc_blind/`.

Blinded protocol. The audit drew a seeded (20260801), stratified sample of 60 distinct cited names over $\text{arm} \times \text{oracle-verdict}$ strata (per arm: 6 flagged hallucinated, 5 exists, 1 renamed). The grader saw only the blinded sample (name + one context line, shuffled, no verdicts) and raw `git grep` evidence at the pinned trees (61a5e4f338, cross-checked at 9944fe2973), wrote truth labels, and only then unsealed the key. Grading required namespace resolution, `to_dual`-generation tracing, `_root_.` declarations, and structure-field projections — exactly the care an exact-string oracle cannot apply.

Truth labels: real 46 · nonexistent 4 · not_a_citation 9 (extractor noise) · renamed 1.

Confusion (oracle $\rightarrow$ truth):

	truth real/renamed	truth nonexistent	truth not_a_citation
oracle hallucinated (30)	18	4	8
oracle exists (25)	24	0	1
oracle renamed (5)	5	0	0

Binary scores (positive = hallucinated): strict precision $4/30 = \mathbf{13.3\%}$ [5.3, 29.7], recall $4/4 = 100\%$ [51.0, 100]; lenient (noise rows excluded) precision 18.2% [7.3, 38.5]. None of the oracle’s 30 negative verdicts (25 *exists* + 5 *renamed*) was a missed fabrication (FN=0); read as verdicts rather than a pooled negative class, 24/25 *exists* names denoted real declarations (one was extractor noise) and 4 of the 5 *renamed* verdicts were in truth real names.

The four confirmed fabrications (zero-hit at both revs), all from arms without formal tools: `FractionField` (A; real name `FractionRing`), `Localization.At` (B; `Localization.AtPrime`), `Ideal.vanishingLocus` (B; `zeroLocus/vanishingIdeal`), `Nat.divisorSum` (B; `ArithmeticFunction.sigma`).

The 26 false flags decompose into six mechanical modes: namespace short names standard under `open` (13), self-declared theorem headers (4), dot-notation on a variable (5, e.g. `M.det`), comment prose (2), an import-line module name (1), a mid-dot-chain fragment (1).

Repair rules R1–R5: drop comment tokens, import-line names, self-declarations, and single-letter dot-notation heads; classify a name as existing if some single-segment prefix `NS.` makes it an indexed declaration. Validation: the repaired classifier agrees with the blinded truth on **59/60** (sole miss: the mid-dot-chain fragment `Laurent.coeff.support`).

Run-level rates, both instruments (n=100/arm; citation-level shown for context — citations cluster within runs, so run-level carries the inference):

arm	raw runs ≥ 1 halluc	raw citation-level	repaired runs ≥ 1	repaired citation-level	R5 reclassified	R1–R4 dropped
A	54/100 [44.3, 63.4]	94/443 = 21.2%	37/100 [28.2, 46.8]	50/428 = 11.7%	29	15
B	48/100 [38.5, 57.7]	76/429 = 17.7%	30/100 [21.9, 39.6]	38/416 = 9.1%	25	13
C	49/100 [39.4, 58.7]	108/517 = 20.9%	11/100 [6.3, 18.6]	26/500 = 5.2%	65	17
D	23/100 [15.8, 32.2]	32/472 = 6.8%	6/100 [2.8, 12.5]	7/457 = 1.5%	10	15
E	49/100 [39.4, 58.7]	107/505 = 21.2%	10/100 [5.5, 17.4]	14/483 = 2.9%	71	22

Raw run-level McNemar: D-vs-E $p=1.6e-4$, D-vs-C $p=6.9e-5$ (run-level ratio $2.13\times$, not the citation-level $3.1\times$). Repaired: D-vs-E $p=0.454$, D-vs-C $p=0.302$, D-vs-A $p=1.23e-7$, D-vs-B $p=3.9e-5$, E-vs-C $p=1.0$. Commit-clustered bootstrap versions of the significant contrasts (`v3_gate_fixes.json`): D-vs-A -0.310 $[-0.446, -0.193]$ $p=0.0002$, E-vs-A -0.270 $[-0.396, -0.155]$ $p=0.0002$; the null D-vs-E / D-vs-C stay null (clustered $p=0.44$ / 0.31). The R5 column is the citation-style artifact quantified: C/E write namespace-short names (65/71 reclassifications), D copies fully-qualified names out of tool payloads (10).

Held-out revalidation of the repaired classifier (seed **20260802**). The five rules were formulated after the 60-name audit unsealed, so 59/60 is an in-sample fit. A second blinded pass (`bench/analysis/halluc_holdout.{py,json,md}`; blinded intermediates in `bench/analysis/holdout_blind/`) sampled **40 distinct cited names disjoint from the 60** (stratified per arm: 4 raw-hallucinated, 3 raw-exists, 1 raw-renamed; shortfalls $\rightarrow$ exists) from the post-repair fresh rows, graded them blind under the same evidence protocol (raw `git grep` at [61a5e4f338](#), cross-check [9944fe2973](#), run outputs for self-declaration/variable resolution), and only then unsealed both instruments' verdicts. Truth labels: real 31 · nonexistent 3 · not_a_citation 5 · renamed 1.

metric (strict, n=40)	raw oracle	repaired classifier
flagged-class precision	3/20 = 15.0% [5.2, 36.0]	3/7 = 42.9% [15.8, 75.0]
recall (true fabrications)	3/3 = 100%	3/3 = 100%
accuracy	57.5%	90.0%
binary agreement (the in-sample 59/60 measure)	23/40	36/40 = 90.0% [76.9, 96.0]

Sensitivity: reading the borderline `OrderedSemiring` (a class removed by the ordered-algebra refactor; labeled renamed on the `Basis` $\rightarrow$ `Module.Basis` precedent) as nonexistent gives repaired precision $4/7 = 57.1\%$. The verdict: **directionally replicates** — the raw oracle's invalidity reproduces almost exactly ($13.3\% \rightarrow 15.0\%$ precision), the repair transfers most of its value (FPs on sampled flags $17 \rightarrow 4$, accuracy $57.5\% \rightarrow 90.0\%$, zero FNs for either instrument) — but the in-sample fit was optimistic ($98.3\% \rightarrow 90.0\%$ binary agreement, Fisher $p=0.15$), and a repaired flag is roughly a coin toss, not a confirmation. The four residual FP modes are ones R1–R5 never saw: multi-segment namespace short names (`IsZero` $\leftarrow$ `CategoryTheory.Limits.IsZero`), dot-notation on a namespaced def value (`MeasureTheory.volume.restrict`), a multi-char local variable (`M001`), and the removed-class borderline (`OrderedSemiring`). Implications: the adjusted per-arm rates likely overstate true hallucination by roughly $2\times$ and should be read as **upper bounds**; two of the four residual FP modes are namespace-style-correlated (both held-out instances land in C/E-style short-name arms), so the residual bias still runs *against* the free-text tool arms and cannot manufacture D's advantage; and all three held-out confirmed fabrications (`Subgroup.torsion`, `Localization.map`, `Module.GeneralizedEigenspace`) sit in the no-formal-tools arms A/B, as in-sample.

S4 Fresh-set exposure strata (full)

The source is `bench/analysis/fresh_exposure.{py,json,md}` (original, snapshot basis), recomputed on the post-repair rows by `bench/analysis/v3_gate_fixes.{py,json}` §1. Pinned tree [61a5e4f338](#) (content $\sim$ 2026-07-10), the rev C/E's file tools read. Outcomes below are the **post-repair rows** (E's 31 outage rows replaced by their 2026-07-27 reruns) on the **raw** instrument; the superseded snapshot-basis table (E's 429 rows counted as failures) is preserved in `fresh_exposure.md`.

Exposure counts (of 100): full dotted name in-tree verbatim **37**; basename as a declaration header anywhere **64**; basename as header in the task's own module **51** (the stratum basis); gold's commit an ancestor of the pin **49**. (Two tasks are exposed despite post-pin merge commits — measured on tree bytes, not merge metadata.)

arm	exposed (n=51)	unexposed (n=49)
A	9/51 = 17.6% [9.6, 30.2]	11/49 = 22.4% [13.0, 35.9]
B	8/51 = 15.7% [8.2, 28.0]	14/49 = 28.6% [17.8, 42.4]
C	11/51 = 21.6% [12.5, 34.6]	14/49 = 28.6% [17.8, 42.4]
D	24/51 = 47.1% [34.1, 60.5]	18/49 = 36.7% [24.7, 50.7]
E	15/51 = 29.4% [18.7, 43.0]	15/49 = 30.6% [19.5, 44.5]

McNemar by stratum, post-repair: D-vs-E exposed $14/5$ $p=0.064$, unexposed $11/8$ $p=0.648$; D-vs-C exposed $18/5$ $p=0.0106$, unexposed $10/6$ $p=0.454$. The leak's direction favors C/E. **A correction to earlier editions:** the snapshot-basis version of this table showed D's edge over E strongest in the *unexposed* stratum ($17/3$, $p=0.0026$), and v2 concluded that D's advantage was “strongest exactly where there was nothing to leak.” That pattern does not survive the E repair — 24 of E's 31 outage rows fell in the unexposed stratum, so the stratum contrast was outage-driven. On the repaired rows D-over-E, like D-over-C, is *larger in the exposed stratum* (RD $+0.18$ vs $+0.06$) and is a null unexposed; neither stratum is significant alone. The

merge-date split reproduces the repaired pattern (before-pin 14/5 $p=0.064$; after-pin 11/8 $p=0.648$), and the repaired-instrument sensitivity gives the same qualitative picture (D-vs-E exposed 15/8 $p=0.21$; unexposed 13/9 $p=0.52$). The judge-endpoint exposure split is §S11.4.

S5 Turn-budget sensitivity

The source is `tier1_reanalysis.md` §5, corrected to the post-repair rows by `v3_gate_fixes.py` §4. E overran the advisory 30-turn budget on **48/100** repaired rows — 16 of the 31 rerun rows plus 32 of the 69 originals; the earlier as-run count of 32 treated E’s 31 outage rows (turns=1) as within budget, which also inflated the old within-budget count (E 68) and the old $n=45$ pair analysis. Corrected per-arm within-budget run counts: A 100, B 100, C 50, D 62, E 52. Restricted to pairs where both arms stayed within the advisory 30 turns (raw instrument, post-repair rows): D-vs-E $n=35$, 51.4% vs 40.0%, 7/3 discordant, $p=0.34$; D-vs-C $n=35$, 54.3% vs 25.7%, 11/1, $p=0.0063$; D-vs-A $n=62$, 46.8% vs 25.8%, 19/6, $p=0.0146$. Consistent in direction; for D-vs-E, inconclusive within budget.

S6 Tool-call census, cost/latency, and figure 5

Tier-1 eval-split trace attribution (`bench/trace_analysis.py`): 98–100% of traced tool-arm runs cite at least one declaration that surfaced in a tool result; 35% of arm-D eval runs cite a declaration from a `brain_bridge/brain_transfer` result. Among *hallucinated* citations (raw oracle), the checked-and-cited-anyway rate on eval rows is 35% (C), 30% (E), 13% (D); on the fresh split — the paper’s primary evidence base — the same statistic is C 44%, D 34%, E 61%: verify-before-cite discipline transferred imperfectly to the held-out set, and worst in E.

v2 census (figure: `figures/fig5_tooluse.pdf` in the typeset build, regenerated on the repaired run rows; the per-tool counts quoted here are the as-run census — the race repair raises only F’s means, since its 190 condemned rows had zero calls; basis stated per clause): W averages 3.5 calls/run on QR-810 and 10.6 on MPR, ≈ 4.0 pooled (QR-810 counts: `decl_exists` 1,251 + `brain_bridge` 608 — verify-then-cite made mechanical; pooled QR+MPR counts: `decl_exists` $\sim 1,473$ + `brain_bridge` ~ 727); WF 4.6 pooled QR+MPR in a genuine dual-toolkit mix (`decl_grep` $\sim 1.5k$, `decl_exists` $\sim 0.9k$, `brain_bridge` $\sim 0.7k$, `decl_read` $\sim 0.5k$, `loogle` $\sim 0.4k$); repaired F averages 2.8 (QR) and 8.2 (MPR) calls/run. Under the manual, `brain_cell` misuse (63.3% failure over 482 W-arm calls) falls to a single call in WF. Same-eval-set caveat as the manual (§S8).

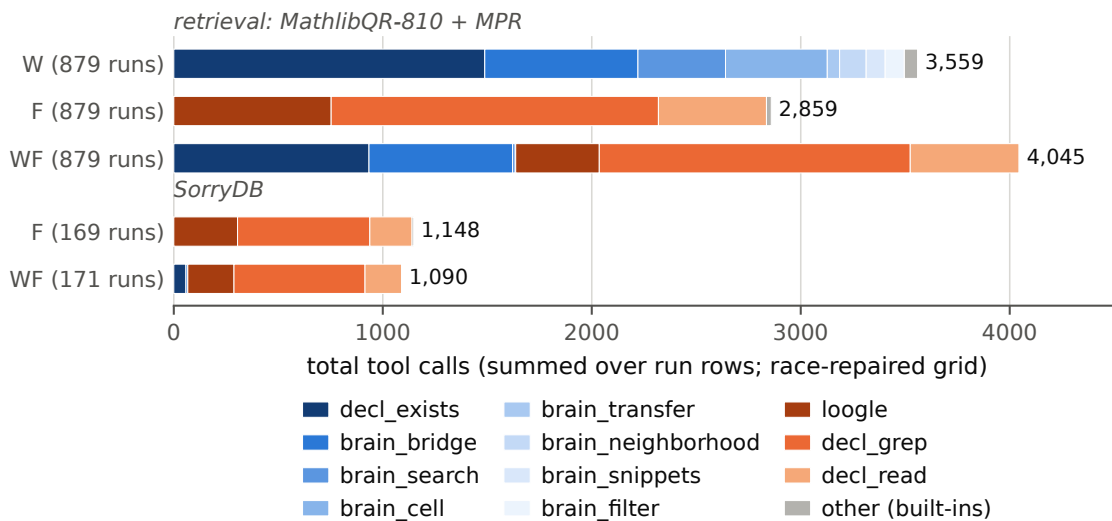


Figure 1: The v2 tool-use census by tool family (Brain blues, formal oranges), summed over run rows — regenerated on the race-repaired grid (`fig_f4_tooluse.py`; census assertions against the §S6 values inside the script). The race repair raises only F’s totals; the prose census above quotes the as-run figures, basis stated per clause.

The cold-start-race repair (grid provenance). The §3.4 race was found by the U launch and repaired across seven commits. 458891dc built the U arm (bare $W \cup F$ union, no manual); its first launch showed 13/49 rows with both MCP servers still **pending** at the init event and zero tool calls, and 834a130a taught the runner to capture that init signature, condemn-and-retry such rows, stagger the first wave, and raise the attach timeout. 99a2075f byte-archived the 194 affected grid rows (F: 175 QR-810 + 15 MPR; W: 2 + 2; manifest of qids included) to `bench/v2/runs/agent/race_condemned_archive/`; f041928a rerun all 194 attach-verified in place under the fixed harness (the 4 stubborn W rows against the original fast-attaching local worker — the asymmetry that had spared W), leaving **zero race rows in every arm $\times$ benchmark**; N/U/WF audit race-free before and after (N has no MCP by design), and pre-existing format/timeout failures (N 143+15, F 1, W 5, WF 1) are untouched. c1dea98f landed the 879 attach-validated U rows (26 condemned first attempts rerun; \$187.20); 322e84f5 and dd3eb689 are the union-ablation and repaired-grid analyses (`union_ablation.{py,json,md}`, `grid_repaired.{py,json,md}`). Repair cost \$33.08. Scores moved only for F (QR R@10 .8309 $\rightarrow$.8457, nDCG .7901 $\rightarrow$.8089; MPR gR@10 .4532 $\rightarrow$.5468); N/U/WF rows are byte-identical and W’s aggregates are unchanged at four decimals.

Per-query cost/latency (QR-810) — repaired grid, recomputed from the final run rows (as-run table: `retrieval_repair.md` §3; CLI cost estimates under Max auth):

system	mean \$/query	mean wall s	median wall s	mean calls	mean turns
N	0.085	15.1	8.9	0.0	1.0
F	0.143	13.2	10.6	2.8	3.8
W	0.197	23.7	11.2	3.5	4.3
U	0.197	12.4	9.8	3.1	4.1
WF	0.207	15.2	12.5	4.2	5.2
retriever-mode wikibrain	~ 0	—	—	1	—

MPR (repaired grid): N \$0.185/50.7s/0 calls · F \$0.370/38.8s/8.2 · W \$0.459/62.6s/10.6 · U \$0.403/33.1s/7.9 · WF \$0.418/38.2s/9.3. U-arm totals: \$159.37 (QR-810) + \$27.84 (MPR) = \$187.20.

N format-repair detail (`retrieval_repair.md`): symmetric tiered extraction (relaxed JSON $\rightarrow$ back-ticked names $\rightarrow$ bare dotted identifiers $\rightarrow$ compound tokens) over all assistant text, identically for every arm’s empty rows. QR: N 143 empty rows $\rightarrow$ 104 repaired ≥ 1 name but only 12 recover the gold; strict 0.6333 $\rightarrow$ lenient 0.6481; oracle ceiling 0.8099. F/W/WF have 1/5/1 empty rows on the repaired grid (2/5/1 as-run) and move ≤ 0.0012 . The five W empties are 420s hard timeouts, not format failures. MPR: N 15 empty $\rightarrow$ 1 gold recovered (0.2025 $\rightarrow$ 0.2062). Per-tool provenance detail (surfaced-by-tool, first-written-in, per-style tables): `bench/analysis/retrieval_provenance.md` — an as-run trace pass; its F row counts the 175 race rows’ hits as memory (main §5 caveat).

S7 Snapshot rot: the detailed narrative

The original SorryDB freeze (8 repositories, 164 tasks) was prepared against sorrydb.org’s SorryDB-2601 evaluation split with pinned upstream commits. At verification time — roughly six months after the pins were minted — 74 of 164 tasks (45%) were unreproducible: the pinned commits of GlimpseOfLean, LeanCourse25, and most Foundation and SemicircleLaw tasks no longer exist on GitHub, their branches force-pushed or rebased away. The rot is invisible to any cheap check: GitHub’s archive endpoint answers a preflight HEAD request for a dead commit with HTTP 200, and only an actual fetch fails. The refreeze (`bench/v2/sorrydb_prep.py`, commit db679e8d) keeps verified-fetchable pins only — every task’s repository archive downloaded and its splice point confirmed — yielding 171 tasks across 10 repositories. Lesson for benchmark builders: a pin is not a snapshot; archive the bytes, not the reference, and verify by fetching, not by HEAD.

S8 The WF agent manual (verbatim) and its timeline

Timeline (verified commit times). N/F/W results committed 22:27 on 2026-07-24; the manual 00:30 on 2026-07-25; WF results 01:15 on 2026-07-25. The manual distills measurements from the same evaluation queries WF was then scored on — its bracketed figures (per-style nDCG, the 63% brain_cell failure rate, the 143/810 format failures, call counts) are the N/F/W/system-mode eval-set aggregates, reproduced exactly.

WF is therefore development evidence of a maximally equipped, briefed agent, not an untouched test result. (v1 also misattributed the 63% figure to “the Tier-1 traces”; it is the v2 W-arm figure — 482 calls, 63.3% failed.)

bench/v2/AGENT_MANUAL.md, prepended verbatim to every WF-arm prompt; a **post-hoc, benchmark-informed** artifact.

Tool manual: searching Mathlib with the Brain + formal search

You have two complementary toolkits. Every claim in this manual is measured from ~5,500 logged tool calls across 3,500 benchmark runs (Tier-1 + Bridge v2); numbers in brackets are those measurements.

The mental model

- **The Brain (wikibrain, 8 tools)** is a *curated knowledge graph* joining informal mathematics (Wikipedia/Wikidata concepts, external databases) to Mathlib declarations. It knows what things *mean* and how concepts relate — but it only “atomizes” declarations that some concept, annotation, or tag claims (~12k of Mathlib’s declarations). It is a map, not the territory.
- **Formal search (3 tools)** operates on *all* of Mathlib directly: type-pattern search (loogle), source grep, source reading. It knows everything that exists but nothing about what it means informally.

Rule of thumb: **route by what you’re holding**. Holding a concept described in words → start with the Brain. Holding a type shape or name fragment → start with formal search. Always finish with `decl_exists` on every name you intend to output.

Formal search tools

loogle

Type-pattern and constant search (the public loogle.lean-lang.org service).

- USE for: “a lemma whose statement looks like `_ * _ ≤ _ * _`”, searches by involved constants (`Real.sqrt`, `tsum`), name substrings.
- Strengths: covers ALL of Mathlib; ~0% call errors [578 calls, 0 errors]; the best premise-finding tool — the loogle+grep toolkit matched a specialist premise retriever [group-recall@10 0.453 vs 0.461 SOTA].
- Weaknesses: needs Lean syntax fluency; a wrong pattern silently returns unrelated hits; small result payloads mean you often need `decl_read` next.

decl_grep

Regex search over the Mathlib source checkout (ripgrep).

- USE for: name fragments (“cantor”), notation, docstring phrases, finding the file a declaration lives in.
- Strengths: the workhorse [1,206 calls, 0% errors]; catches things loogle’s type index can’t (comments, docstrings, naming conventions).
- Weaknesses: lexical only — if the concept’s name doesn’t appear in source, grep cannot find it (descriptive queries like “special case of X” score worst for grep-based agents [nDCG 0.384 vs the Brain’s 0.523]).

decl_read

Read declaration source (with surrounding context) from the checkout.

- USE for: verifying a candidate actually says what you think BEFORE citing it; getting exact signatures and hypotheses.
- Strengths: ground truth for any decl in Mathlib [397 calls, 0% errors].
- Weaknesses: you need the file/name first — it is a confirmation tool, not a discovery tool.

Brain tools

brain_bridge — THE ENTRY POINT for informal→formal

Free text in, existence-verified declarations + owning atoms out, with signatures, import lines, and one-hop dependency context.

- USE for: “the concept described as ⟨words⟩” — especially nicknames and special-case descriptions, where the Brain’s curated bonds beat grep [special_case nDCG 0.523 vs 0.384; nickname 0.779 vs 0.757].
- CRITICAL LIMITATION (measured): its free-text matching is label/alias anchored, NOT semantic. A single call with a descriptive paraphrase usually misses [single-shot benchmark: 3.6% R@10 vs 82% for an agent that reformulates]. **Never accept one miss as the answer: reformulate 2–4 times** — try the concept’s canonical name, a synonym, the head noun alone. 8.9% of calls return match:“none” [727 calls]; that response includes nearest-candidate suggestions — read them.
- The `hits` list is existence-verified; bond quality matters: **exact** means the decl IS the concept; **formalizes**/weaker bonds are nearby.

brain_search

Fuzzy label + alias search returning atoms (cells).

- USE for: resolving a name you half-know into an atom id; browsing what the Brain calls something [400 calls, 0% errors — the reliable fallback when `brain_bridge` misses].
- Weaknesses: returns atoms, not ranked decls — follow with `brain_cell` on the winning atom.

brain_cell

The full atom card: Lean code, Wikidata description, DB snippets, organs.

- USE for: everything known about ONE object after bridge/search resolved it.
- **THE #1 MISUSE [63% of 482 calls failed]: calling it with a bare `decl:Mathlib:Name` for an arbitrary declaration.** The Brain only has atoms for concept-joined decls; “unresolvable key” means *no atom owns this decl*, NOT that the decl doesn’t exist. For arbitrary decls use `decl_exists` (works for all of Mathlib) and `decl_read` for source. Call `brain_cell` only with ids that came OUT of a Brain tool.

decl_exists — *THE DISCIPLINE TOOL*

Batch existence check for declaration names (backed by the full oracle — covers ALL of Mathlib, unlike `brain_cell`).

- **USE ALWAYS, on every name you are about to output.** This is the measured difference between grounded and hallucinated citations: agents that verified before citing cut hallucinated-but-cited-anyway to 13% vs 33% for agents relying on fuzzy search results alone. It is cheap [1,473 calls, 0.1% errors] and batched — check 10 names in one call.
- Also returns verified-rename suggestions for stale names.

brain_neighborhood

The synapse graph around an atom (depends/links/relates, cursored).

- USE for: “what connects to X” — finding the connecting lemma between two concepts, walking dependencies.
- Weaknesses: needs a valid atom id [24.8% of calls errored — same unresolvable-key trap as `brain_cell`; resolve the atom first].

brain_transfer / *brain_snippets* / *brain_filter*

Narrower tools: label↔decl jumps for KNOWN labels (transfer), stored source snippets (snippets), facet enumeration (filter). High misuse rates when fed non-atom ids [40%/57% errors]. Prefer `brain_bridge`/`brain_search` entry; reach for these only on ids the Brain already returned.

Routing playbook

You are holding	Do this
A concept in words	<code>brain_bridge</code> → (miss?) reformulate → <code>brain_search</code> → <code>brain_cell</code>
A special case / nickname	<code>brain_bridge</code> (the Brain’s best terrain)
A type shape	<code>loogle</code>
A name fragment	<code>decl_grep</code>
Premises for a proof	<code>loogle</code> + <code>decl_grep</code> FIRST [0.453 vs 0.272], Brain for concept grounding
A candidate name to cite	<code>decl_exists</code> (batch, always) → <code>decl_read</code> if the statement matters
An atom id from any Brain tool	<code>brain_cell</code> / <code>brain_neighborhood</code> freely
A bare decl name needing info	<code>decl_exists</code> + <code>decl_read</code> — NOT <code>brain_cell</code>

Anti-patterns (each one measured in prior runs)

1. **Citing unverified names.** 33% of hallucinated citations came from agents that had search evidence in front of them and cited a wrong name anyway. `decl_exists` is binary and kills this.
2. **One query, then giving up.** The single-call miss rate on descriptive queries is ~96%; agents that reformulate reach ~82%. Iteration IS the algorithm.
3. **`brain_cell` as a generic decl inspector.** 63% failure rate. Brain atoms ≠ all of Mathlib.
4. **Answering outside the requested format.** 143/810 no-tool runs were scored 0 for exactly this. Re-read the output contract before replying.
5. **Inventing tool names** (e.g. `logue`, `lookie`). Use the names above.

(*End of verbatim manual.*) Its bracketed measurements are the as-run pre-repair aggregates as WF saw them — e.g. the `loogle+grep` premise figure it quotes as 0.453 is 0.547 on the repaired grid (§S6) — and are preserved verbatim, not corrected.

S9 SorryDB: full bookkeeping

The frozen split holds n=171 tasks/arm across 10 repositories, and all 203 candidate verdicts are definitive — 183 kernel pass/fail plus 19 unspliceable and 1 verify-timeout that never reached the kernel (commit 9682b3b1; the 8 curve25519 stragglers verified as 0 proved). `verify.jsonl` carries 2 additional off-frozen N rows verdicted `env_broken` (205 lines total), correctly excluded from the 203.

	N no tools	F formal	WF union+manual (post-hoc)
run rows present	168	169	171
no output (timeout / no lean block)	70	15	11
honest give-up (literal <code>sorry</code>)	40	83	86
candidate proofs submitted	58	71	74
proved (kernel)	2	9	10
failed	50	55	57
unspliceable	5	7	7
verify timeout	1	0	0
proved / 171	1.2% [0.3, 4.2]	5.3% [2.8, 9.7]	5.8% [3.2, 10.4]
repo-clustered boot 95% CI	[0.0, 2.7]	[0.0, 10.7]	[0.0, 12.2]
proved / candidates	3.4%	12.7%	13.5%
total cost (USD)	58.97	102.76	108.87
cost per proved	29.48	11.42	10.89
mean wall-clock / task	120 s	198 s	183 s

3 N and 2 F tasks lack run rows (runner losses). Distinct sorries proved: 13; N’s are a strict subset of F’s and WF’s; $WF \cap F$ 6, WF-only 4, F-only 3. Per-repo: PersistentDecomp N1/F3/WF3, LeanAPAP 0/4/5, MiscYD 1/2/2, the other seven repos 0 everywhere. $WF - F$: +0.0058, repo-clustered 95% CI [+0.0000, +0.0185] (the lower endpoint is exactly 0 because $WF \geq F$ in every repo; 34% of resamples show no difference or worse); task-level exact McNemar 4/3, $p=1.0$. Cost-per-proved is bookkeeping only (unstable at 2/9/10 successes). The published SorryDB agentic best (~30%) is not comparable: different task subset, different protocol (published agents iterate against the build; ours drafted one-shot).

S10 Per-arm cost tables

Tier-1 campaign means per task (all 471 runs/arm as run, pre-repair; E’s 31 error rows cost ≈ 0): A \$0.034 · B \$0.048 · C \$0.140 · D \$0.121 · E \$0.128; mean wall-clock C 116 s, D 89 s, E 106 s. tokens-to-solve was never computed, so P2 is untested; v1’s “cheaper and better” framing remains withdrawn. **Judge pass:** \$81.86 (CLI-reported) for 500 gradings + the 50-item re-grade. **v2 retrieval:** per-query tables in §S6. **SorryDB:** totals in §S9.

S11 Blind LLM-judge grading: full tables

Sources are `bench/analysis/judge_fresh_run.py` and `judge_fresh_summary.{py,json,md}`, with the repaired-leg conjunction from `bench/analysis/conjunction_repaired.{py,json}` (deterministic — no sampling). Judge `claude-sonnet-5`; blind (informal statement + gold with binders + candidate; no arm identity, no tools, empty cwd); 500/500 rows, 0 errors; no-output rows auto-fail. **Uncalibrated; exploratory.** The judge ran before the oracle repair, so the conjunction is reported under both typecheck instruments; the judge leg is identical in both.

S11.1 Fresh-100 rates

Strict equivalence (same proposition, same hypotheses):

arm	k/n	Wilson 95% CI
A	11/100	[6.2, 18.6]
B	9/100	[4.8, 16.2]
C	43/100	[33.7, 52.8]
D	8/100	[4.1, 15.0]
E	46/100	[36.6, 55.7]

Evaluated equivalence (mathematical equivalence, high confidence): A 17/100 [10.9, 25.6] · B 16/100 [10.1, 24.4] · C 51/100 [41.3, 60.6] · D 19/100 [12.5, 27.8] · E 53/100 [43.3, 62.5].

Conjunction (grounded-typecheck $\wedge$ evaluated), both instruments:

arm	raw leg	repaired leg [Wilson 95]
A	8/100	8/100 [4.1, 15.0]
B	9/100	9/100 [4.8, 16.2]
C	16/100	20/100 [13.3, 28.9]
D	15/100	16/100 [10.1, 24.4]
E	18/100	21/100 [14.2, 30.0]

Eight rows flip under the repair (C 4, E 3, D 1 — the repaired-clean typecheck passes that the judge also evaluated equivalent); B/A are essentially untouched, and between-tool parity is preserved.

S11.2 McNemar, fresh-100

Evaluated equivalence: D-vs-E 1/35 $p=1.1e-9$ · D-vs-C 3/35 $p=6.7e-8$ · D-vs-A 7/5 $p=0.774$ · E-vs-A 37/1 $p=2.8e-10$ · E-vs-B 39/2 $p=7.8e-10$ · E-vs-C 13/11 $p=0.839$.

Conjunction, raw leg: D-vs-E 6/9 $p=0.607$ · D-vs-C 8/9 $p=1.0$ · D-vs-A 9/2 $p=0.065$ · E-vs-A 11/1 $p=0.0064$ · E-vs-B 12/3 $p=0.035$ · E-vs-C 6/4 $p=0.754$.

Conjunction, repaired leg (`conjunction_repaired.py`): D-vs-E 5/10 $p=0.302$ · D-vs-C 8/12 $p=0.503$ · D-vs-A 10/2 $p=0.039$ · E-vs-A 14/1 $p=0.00098$ · E-vs-B 14/2 $p=0.0042$ · E-vs-C 6/5 $p=1.0$. The only classification change at $\alpha=.05$ is D-vs-A ns $\rightarrow$ sig — a strengthening of tools-versus-none; every between-tool conjunction contrast stays null under both instruments.

Commit-clustered bootstrap versions (44 clusters, B=10,000; `v3_gate_fixes.json`): evaluated equivalence D-vs-E -0.340 $[-0.462, -0.220]$ $p=0.0002$, E-vs-A $+0.360$ $[+0.233, +0.469]$ $p=0.0002$; conjunction (repaired leg) D-vs-A $+0.080$ $[+0.011, +0.152]$ $p=0.037$, E-vs-A $+0.130$ $[+0.055, +0.226]$ $p=0.0002$, D-vs-E -0.050 $[-0.143, +0.021]$ $p=0.247$. No classification changes: everything significant unclustered survives clustering, and the nulls stay null.

S11.3 Completed-69 continuity

Evaluated: A 10/69 (.145) · B 9/69 (.130) · C 37/69 (.536) · D 11/69 (.159) · E 37/69 (.536); D-vs-E 0/26 $p=3.0e-8$; conjunction (raw leg): C 8/69 (.116) · D 10/69 (.145) · E 9/69 (.130), D-vs-E 4/3 $p=1.0$, D-vs-A 7/0 $p=0.016$; conjunction (repaired leg): C, D, and E at exactly 11/69 (.159) each, D-vs-E 4/4 $p=1.0$, D-vs-A 8/0 $p=0.0078$.

S11.4 Exposure split (evaluated equivalence)

arm	exposed (n=51)	unexposed (n=49)
A	8/51 (.157)	9/49 (.184)
B	9/51 (.176)	7/49 (.143)
C	36/51 (.706)	15/49 (.306)
D	10/51 (.196)	9/49 (.184)
E	35/51 (.686)	18/49 (.367)

D-vs-E exposed: 1/26, $p=4.2e-7$. D-vs-E unexposed: 0/9, $p=0.0039$ — the inversion narrows off-leak but does not vanish; E produces more judge-equivalent statements than D even where the gold was not retrievable. The conjunction (S11.2) is where parity appears. This is reported as measured; the human queue below is how it gets adjudicated.

S11.5 Self-consistency and the human queue

Fixed 50-item seed-stratified re-grade (seed 20260727, 10/arm): strict agreement 98.0% (one flip), evaluated agreement 100.0%. Human grading queue = the 36 D/E-discordant tasks on evaluated equivalence (they drive the D-vs-E McNemar): fresh_002, 004, 006, 007, 011, 012, 013, 014, 019, 021, 023, 026, 027, 032, 044, 048, 049, 050, 051, 056, 057, 059, 062, 063, 064, 067, 069, 076, 077, 085, 086, 092, 094, 095, 096, 099.

S12 Responses to the external reviews

Both reviews are preserved verbatim in `docs/research/review/` (REVIEW-1.md, REVIEW-2.md), with our independent claim-by-claim verification of review 1 in `verification-of-review-1.json`.

S12.1 Review 1 (of v1, 2026-07-27) — actions taken in v2

All eight concerns confirmed in substance (29 claims confirmed, 1 partial, 1 refuted). Two factual claims did not survive: MathlibMPR does not cluster by PR (69 tasks, 69 distinct PRs), and arm E’s 31 declaration-less runs were contiguous 429 errors, not interface overload — the second correction removed both the review’s overload reading and v1’s error-inflated headline $p=2.4e-5$.

#	Concern	Action
1	“Success” measures no semantic correctness	renamed grounded typecheck rate; judge pass; calibration owed
2	D vs E does not isolate the join	“join carries the effect” retired; bundled-condition language; the 2×2 factorial has since been run (v3.1, §4.4/§S15): both main effects null
3	Arm E failed its manipulation check	disclosed; yoked control on the roadmap
4	Fresh set not isolated from formal-search sources	renamed post-Brain-index; exposure strata; frozen snapshots on the roadmap
5	WF is test-set tuned	labeled post-hoc, benchmark-informed everywhere; provenance + timeline (§S8); bare-union arm launched
6	Independence and execution problems	declaration-clustered reanalysis; CIs throughout; disclosures
7	Preregistered experiment incomplete; deviations silent	execution inventory (§S1); register labels; zero-confirmatory up front
8	SorryDB evidence thin	all 203 verdicts completed; WF-vs-F indistinguishable; cost claim deleted

S12.2 Review 2 (of the v2 draft, 2026-08-01) — actions taken in v3

#	Recommendation	Action in v3
1	Lead with repaired full-100; one coherent paired inference	commit-clustered paired bootstrap is the only main-text framework; completed-69 and Wald/McNemar mismatch moved here (§S2)
2	Cluster the fresh set by commit/family	done (fresh_clustered.py): 44 commits, 57 files, 59 families; all levels + collapses (§S2.3); D–E widens exactly as predicted
3	Human semantic evaluation	full 500-row blind judge complete with self-consistency; 36-task discordant queue defined; human grading and calibration still pending — main §4.2 stays exploratory
4	Factorial ablation	DONE in v3.1: the preregistered 2×2 (prereg 3658bd58, before any row) is run and analyzed (§4.4/§S15) — join +0.030 p=0.645, verifier +0.050 p=0.230, interaction −0.100 (redundancy); the bare-union U ablation had already shown the union inert (U–F null everywhere), the WF gain manual-driven on QR, and formal tools carrying MPR
5	Characterize the Brain; retrieval provenance decomposition	done (brain_artifact.py , main §3.1; retrieval_provenance.py , main §5)
6	Real related work (CRAMF, Aria, +)	done; every citation verified against arXiv on 2026-08-01 (docs/research/review/related_work_notes.md); comparison table in main §2
7	Correct two overclaims (memorization; “clean verifier finding”)	memorization → benchmark-familiarity language (main §4.1); verifier finding restated after blinded oracle validation — between-tool contrast dissolves under the repaired instrument (main §4.3), DDR cited as convergent evidence
8	Repair the retrieval evaluation	symmetric lenient extraction for every arm, cost/calls columns, retriever-vs-agent blocks, Brain-gold coverage, provenance decomposition (main §5, §S6); cold-start-race grid repair + U ablation complete; WF label kept
cuts	move manuals, inventories, strata, response tables, file maps out	this supplement is that move

S13 Changelog

- **v1 (2026-07-25)**. Initial report. Headline claims later retired: “the join carries the effect” (p<0.0001 driven by an unrecognized outage), “42% success” (no equivalence leg), “contamination-proof fresh set”, WF as a SOTA row, cost-per-success claims.
- **v2 (2026-07-31)**. Post-review-1 corrected edition: metric renamed grounded typecheck; arm-E 429 block repaired; register labels (preregistered-modified / exploratory / corrective) and zero-confirmatory statement up front; exposure strata; declaration-clustered retrieval statistics; WF relabeled post-hoc; SorryDB verdicts completed; judge and union slots left open.
- **v3 (2026-08-02, this edition)**. Post-review-2 restructure into an 8-page paper + this supplement. New evidence: blinded hallucination-oracle validation (13.3% precision) + 5-rule repaired instrument, with all Tier-1 headline numbers recomputed under it; commit-clustered bootstrap as the single inferential framework; completed blind judge pass (500/500) with the three-layer contamination×endpoint analysis; Brain artifact characterization; retrieval provenance decomposition (verification + routing, not content);

N format-failure repair; the cold-start-race grid repair (194 rows rerun attach-verified; F QR R@10 .831→.846, MPR .453→.547) with the bare-union U ablation and final contrast set; the repaired-leg judge conjunction; verified related work; cost columns; conflict/AI-use disclosures. Still open: human grading of the 36-task queue, judge calibration, the 2×2 factorial.

- **v3 post-gate corrections (2026-08-03).** After an adversarial verification gate: held-out blinded revalidation of the repaired oracle (§S3; flags become upper bounds); §S4 exposure strata and §S5 turn budgets recomputed on the post-repair rows (the “strongest-where-nothing-to-leak” claim retracted as outage-driven; E overruns 48/100); commit-clustered bootstraps extended to every significant §4.2/§4.3 contrast; Tier-1 attach audit; fresh-task provenance paragraph; five-arm judge Layer 1; census bases relabeled; assorted number and wording corrections (681, S6 fresh rates, SorryDB verdict categories, anchor provenance).
- **v3.1 (2026-08-18, this edition).** The preregistered 2×2 join × verifier factorial, run and analyzed exactly per BRIDGE-FACTORIAL.md (prereg 3658bd58 2026-08-07; 400 rows ba35fe7f; new main-text §4.4 + §S15): both preregistered main effects null (join +0.030 p=0.645, verifier +0.050 p=0.230), exploratory interaction −0.100 (redundancy, not synergy); abstract, §1, §7, and §8 updated to the causal answer. One scoring-phase infrastructure incident (REPL server death mid-D’; the whole arm re-typechecked, §S15.1) documented with full cell provenance. Judge-graded secondaries completed same-day after an authentication delay (400/400 + 40-item re-grade, self-consistency 100%/97.5%): the §4.2 contamination-by-endpoint inversion reproduces (evaluated-equivalence JOIN −0.265 p=0.0002, exploratory; conjunction parity, all effects null; §S15.4). Still open: the 36-task human queue and judge calibration.

S14 Data availability and file map

Everything needed to recompute the report lives in the private preservation repository **Deicyde/wikilean-bridge-experiment** (report at the root; supplement under **report/**). Provenance commits below refer to the WikiLean working repository.

Key commits. 0d36f266 preregistration (2026-07-16) · f97e67e4 review-1 preservation + byte-preserved 429 archive · 19a90209 arm-E repair driver (MCP-attach detection) · 64c48052 v2 corrective statistics (tier1_reanalysis, fresh_exposure, retrieval_clustered) · 9682b3b1 SorryDB kernel verdicts complete (203/203) · 36f3a472 report v2 + scored summary v2 (bridge_summary_v2.json) · 662db420 review-2 analyses (commit-clustered inference, blinded oracle validation, N format repair, Brain artifact table, provenance decomposition) · 3cb8fa55 blind judge grading complete (500/500, self-consistency, conjunction tables, human queue) · 9d10e44f repaired-oracle Tier-1 recompute (success_repaired) · 458891dc U-arm harness (bare $W \cup F$ union, no manual) · 834a130a cold-start-race condemnation + staggered launches · c1dea98f U run rows (879, attach-validated) · 322e84f5 union-ablation analysis · 99a2075f race-row archive (194 rows, byte-preserved) · f041928a repaired F/W run rows (zero race rows remain) · dd3eb689 repaired-grid contrasts + refreshed union ablation · 2a9f6b91 REPL routing fix (“honest labels”).

Path (preservation repo)	Contents
docs/research/BRIDGE-EXPERIMENT.md	preregistration incl. deviations 1–7
docs/research/review/	reviews 1–2 verbatim; verification-of-review-1.json; related_work_notes.md (verified citations)
bench/*.py, bench/arms/	Tier-1 harness: runner, scorer, REPL rig, trace analysis, arm manifests
bench/data/bridge_tasks.jsonl	371 ProofNet# tasks (pinned fork, MIT)
bench/data/fresh_tasks.jsonl	100 post-Brain-index tasks + determinacy annotations + holdout check
bench/data/runs/	all 2,355 Tier-1 run rows incl. 31 repaired E rows
bench/data/runs_E_fresh_429_archive/	the 31 original 429 rows, byte-preserved
bench/analysis/bridge_summary_v2.json	scored fresh-500 summary + repair provenance
bench/analysis/tier1_reanalysis.*, fresh_exposure.*, retrieval_clustered.*	v2 corrective statistics
bench/analysis/fresh_clustered.*, halluc_validation.*, success_repaired.*	v3: clustered inference; blinded oracle validation; repaired-instrument recompute
bench/analysis/halluc_holdout.* (+holdout_blind/)	held-out blinded revalidation of the repaired oracle (40 disjoint names, seed 20260802)
bench/analysis/v3_gate_fixes.{py,json,md}	post-gate analyses: post-repair S4 strata, clustered §4.2/§4.3 bootstraps, Tier-1 attach audit, corrected turn budgets, five-arm judge table, fresh-task provenance
bench/analysis/judge_fresh_run.py, judge_fresh/, judge_fresh_summary.*	blind judge pass (500 gradings + re-grade)
bench/analysis/conjunction_repaired.{py,json}	judge conjunction under both typecheck instruments (deterministic)
bench/analysis/retrieval_repair.*, retrieval_provenance.*, brain_artifact.*	v3: format repair; provenance decomposition (as-run trace pass); artifact characterization
bench/analysis/union_ablation.*, grid_repaired.*	U-arm bare-union ablation; race-repaired grid, before/after tables + final contrast set
bench/v2/runs/agent/race_condemned_archive/	the 194 original cold-start-race rows, byte-preserved with qid manifest
bench/v2/ + bench/v2/data/ + bench/v2/runs/	v2 harness, AGENT_MANUAL.md, MathlibQR/MPR (CC BY 4.0), SorryDB freeze, all v2 run rows (repaired grid) with gzipped stream transcripts, verify.jsonl (203 verdicts)
docs/research/BRIDGE-FACTORIAL.md	the 2×2 factorial preregistration (commit 3658bd58, byte-unchanged; empty deviations log)
bench/factorial/	factorial runner (run_factorial.py) + prereg-time live-index census
bench/arms/mcp-{Ep,X,J,Dp}.json	the four factorial arm MCP configs
bench/data/runs_factorial/	all 400 factorial rows + gzipped stream transcripts + conditions.json (commit ba35fe7f)
bench/analysis/factorial_scored.{py,json,md}	Stage-1 mechanical scoring (repaired+raw oracle, fresh-pin typecheck, integrity gate, retc_provenance)
bench/analysis/judge_factorial_run.py	Stage-2 blinded-judge driver + all 400 verdicts + 40-item consistency re-grade (§S15.4)
bench/analysis/judge_factorial/	Stage-3 preregistered analysis (effects, pairwise grid, sensitivity cuts)
bench/analysis/factorial_analysis.{py,json,md}	

Recomputation entry points. Factorial bench/analysis/factorial_scored.py + factorial_analysis.py (runner bench/factorial/run_factorial.py); Tier-1 grading bench/score_bridge.py; repaired instrument bench/analysis/halluc_validation.py (adjusted) + success_repaired.py + held-out revalidation halluc_holdout.py; post-gate analyses v3_gate_fixes.py; clustered inference fresh_clustered.py; exposure fresh_exposure.py; judge judge_fresh_summary.py + conjunction_repaired.py; retrieval bench/v2/score_retrieval.py + retrieval_clustered.py + retrieval_repair.py + retrieval_provenance.py; repaired grid + union ablation grid_repaired.py + union_ablation.py; artifact brain_artifact.py; SorryDB bench/v2/verify_sorrydb.py.

External sources (fetched and verified during the v2/v3 design sweeps): TheoremGraph arXiv:2606.25363 · LeanSearch arXiv:2403.13310 · LeanSearch-v2 arXiv:2605.13137 + github.com/frenzymath/LeanSearch-v2 · LeanExplore arXiv:2506.11085 · SorryDB arXiv:2603.02668 + sorrydb.org · FATE arXiv:2511.02872 · miniCTX arXiv:2408.03350 · LeanDojo arXiv:2306.15626 · LeanAgent arXiv:2410.06209 · Numina-Lean-Agent arXiv:2601.14027 · miniF2F-v2 arXiv:2511.03108 · ProofNet#/ProofNetVerif arXiv:2406.07222 · benchmark-faults survey arXiv:2606.29493 · CRAMF arXiv:2508.06931 · Aria arXiv:2510.04520 · DRIFT arXiv:2510.10815 · DDR arXiv:2511.11990 (the last four verified 2026-08-01, related_work_notes.md).

S15 The preregistered 2×2 factorial (join × verifier): full detail

Preregistration docs/research/BRIDGE-FACTORIAL.md, commit 3658bd58 (2026-08-07) — committed before the harness was built and before any row ran; the file is byte-unchanged since, and its §9 deviations log is empty. Harness commit cbdcf3b7; the 400 run rows (+ gzipped stream transcripts + conditions.json) commit ba35fe7f. Scoring, judging, and analysis are a separate later phase (prereg §4.9), by a different agent from the runner. Artifacts: bench/analysis/factorial_scored.* (Stage 1), bench/analysis/judge_factorial_run.py + judge_factorial/ (Stage 2), bench/analysis/factorial_analysis.* (Stage 3).

S15.1 Arms, execution, and integrity

2×2 over JOIN (Wikibrain joined surface replaces the unjoined wiki+formal stack) × VERIFIER (decl_exists present): E' = arm E's 6 tools; X = E' + decl_exists (7); J = the 7 Wikibrain graph/content tools; D' = all 8 (arm D exactly). All arms: -tools "" (empty built-in set), per-row allowed/disallowed manifests, model claude-haiku-4-5-20251001, CLI 2.1.153, prompt byte-identical to Tier-1, -max-turns 30 (mechanical), formal-search reads pinned to the 61a5e4f338 tree, one interleaved order (seed 20260803 shuffle of all 400 pairs), concurrency 4 with staggered first wave, per-row attach-signature + manifest validation with auto-condemn/retry.

Run 2026-08-08 00:51–03:45 UTC, \$40.91. Integrity gate (re-verified at scoring): 400/400 terminal rows, 0 errors, 0 attach-dirty, 0 manifest mismatches, 0 zero-tool rows, 0 turn-cap violations; per-arm condition hashes uniform and equal to conditions.json. 46 rows hit the mechanical cap (E' 18, X 22, J 0, D' 6) — valid terminal rows scored on their extracted output per prereg §4.2.

Scoring-phase incident (defect-4 class, repaired). The fresh-pin REPL server died mid-pass while typechecking arm D' — during fresh_052's check, whose first verdict recorded the REPL dying — and the 44 subsequent produced rows silently fell back to single-shot bare-environment checks (sub-second failures citing unknown identifiers) — the exact silent-fallback mode the v3 report's §3.4 defect 4 documents. Caught by an elapsed-time audit (all legitimate server verdicts take ≥3s; the other three arms' minima are 3.0–9.8s), repaired by re-typechecking every produced D' row against a restarted, identity-gated server: all 48 healthy-window verdicts reproduced exactly; 23 of the 45 affected verdicts flipped, all False→True (fresh_052 among them). Full cell-level provenance: factorial_scored.json → retc_provenance.

S15.2 Primary endpoint: grounded typecheck (repaired oracle)

arm	cell	k/100	rate	Wilson 95%	raw oracle	typecheck ok	produced
E'	join−, ver−	31	.310	[.228, .406]	26	33	82
X	join−, ver+	41	.410	[.319, .508]	31	42	78
J	join+, ver−	39	.390	[.300, .488]	36	39	100
D'	join+, ver+	39	.390	[.300, .488]	39	40	93

Preregistered effects (commit-clustered paired bootstrap, 44 clusters, B=10,000, seed 20260803, cluster_boot_rd verbatim; α=.05 two-sided):

effect	register	RD	95% CI	p	verdict
JOIN main	confirmatory	+0.030	[−0.093, +0.139]	0.645	null
VERIFIER main	confirmatory	+0.050	[−0.029, +0.121]	0.230	null
interaction	exploratory	−0.100	[−0.245, +0.047]	0.196	no detectable interaction

Six pairwise clustered RDs (supporting descriptives): D'−E' +0.080 [−0.071, +0.207] p=0.310 · D'−X −0.020 [−0.172, +0.120] p=0.840 · D'−J +0.000 [−0.099, +0.080] p=1.000 · X−E' +0.100 [−0.022, +0.215] p=0.127 · J−E' +0.080 [−0.053, +0.202] p=0.259 · J−X −0.020 [−0.155, +0.116] p=0.821.

Raw-oracle sensitivity: rates E' 26 / X 31 / J 36 / D' 39; join +0.090 [−0.006, +0.183] p=0.076; verifier +0.040 [−0.030, +0.103] p=0.298; interaction −0.020 [−0.156, +0.134] p=0.832. The raw instrument flatters the join arms (they copy fully-qualified names from tool payloads; §4.3's citation-style artifact), and the repair removes exactly that artifact — the preregistered primary is the repaired row.

S15.3 Secondary: run-level repaired hallucination (lower better)

Rates: E' 9/100 · X 5/100 · J 11/100 · D' 10/100. Effects: join +0.035 [−0.026, +0.100] p=0.289; verifier −0.025 [−0.081, +0.025] p=0.381; interaction +0.030 [−0.082, +0.142] p=0.666. With formal tools of any kind present, flagged citations no longer separate arms.

S15.4 Secondaries: judge evaluated-equivalence and the conjunction

Graded 2026-08-18, after a same-day authentication delay (the judging CLI's Max OAuth had expired; interactive re-login, then the staged pass ran unchanged): `bench/analysis/judge_factorial_run.py`, protocol identical to the v3 fresh-set judge pass (verbatim `judge_bridge.PROMPT`, gold with binders, arm-substring blinding scan green over all 353 gradeable outputs, no-tools claude-sonnet-5 from an empty scratch cwd, `-max-turns 1`, concurrency 3, 429 wait-retry); the 47 no-output rows pre-decided not-equivalent by definition. 400/400 verdicts, 0 judge errors (three transient max-turns failures re-graded cleanly); judge cost \$52.32. Self-consistency on the fixed 40-item seed-stratified re-grade (seed 20260727, 10/arm): evaluated agreement 100.0%, strict agreement 97.5% (one flip). Both endpoints are preregistered-exploratory; the judge remains uncalibrated (§4.2 caveats apply unchanged).

Evaluated equivalence:

arm	evaluated	Wilson 95%	strict	exposed (n=51)	unexposed (n=49)
E'	46/100	[36.6, 55.7]	39/100	30/51 (.588)	16/49 (.327)
X	42/100	[32.8, 51.8]	36/100	34/51 (.667)	8/49 (.163)
J	20/100	[13.3, 28.9]	13/100	9/51 (.176)	11/49 (.224)
D'	15/100	[9.3, 23.3]	12/100	7/51 (.137)	8/49 (.163)

Factorial effects on evaluated (same clustered machinery): join **−0.265** [−0.368, −0.154] p=0.0002; verifier −0.045 [−0.117, +0.029] p=0.264; interaction −0.010 [−0.152, +0.117] p=0.915. This is §4.2's contamination-by-endpoint inversion reproduced under the factorial's yoked interface: the unjoined arms hold source grep over the checkout containing 51 of the 100 golds and transcribe — their judged-equivalence advantage concentrates on the exposed half and largely vanishes off it — while the join arms' generate-and-verify outputs are grounded but less often judged equivalent.

Conjunction (grounded typecheck ∧ evaluated, the closest faithful@budget analogue): E' 20 / X 25 / J 17 / D' 14 per 100; join −0.070 [−0.161, +0.021] p=0.141; verifier +0.010 [−0.053, +0.065] p=0.822; interaction −0.080 [−0.176, +0.011] p=0.108 — parity: no factor moves the conjunction detectably in either direction.

S15.5 Manipulation checks, descriptives, sensitivity cuts

Verifier usage: X 213 `decl_exists` calls across 94/100 runs; D' 536 across 92/100. Informal-tool touches: E' 2/100 runs, X 4/100 — the unjoined arms again barely open the informal corpus (Tier-1's E manipulation failure, reproduced). Mean assistant turns: E' 20.4 / X 21.4 / J 17.1 / D' 17.4; mean tool calls 30.4 / 30.3 / 26.8 / 26.7. Production: E' 82 / X 78 / J 100 / D' 93 of 100 — the join arms finish inside the 30-turn budget; the unjoined arms cap out (18 and 22 capped rows). The factorial's rates are therefore not comparable to the Tier-1 five-arm table (§4.1): the yoked interface (empty built-ins + mechanical cap + interleaving) is stricter, and D' lands at 39% where Tier-1's D scored 48%.

Sensitivity cuts (primary outcome, exploratory; same machinery):

cut	E'	X	J	D'	join RD [CI] p	verifier RD [CI] p
own-module exposed (n=51)	17	25	20	18	−0.039 [−0.205, +0.108] 0.653	+0.059 [−0.060, +0.170] 0.391
unexposed (n=49)	14	16	19	21	+0.102 [−0.076, +0.243] 0.271	+0.041 [−0.054, +0.130] 0.438
drop 3 live-index leaks (n=97)	31	39	38	38	+0.031 [−0.093, +0.142] 0.632	+0.041 [−0.039, +0.114] 0.353
both-annotator determinate (n=74)	25	35	30	30	+0.000 [−0.156, +0.139] 1.000	+0.068 [−0.013, +0.151] 0.110

Every cut preserves both nulls. The exposed/unexposed join reversal (−0.04 vs +0.10, both null) is the direction §4.2's contamination analysis predicts — the unjoined arms benefit where golds are retrievable — but neither stratum reaches significance.

S15.6 Recomputation

`python3 bench/analysis/factorial_scored.py` (score $\rightarrow$ retc-arm Dp $\rightarrow$ judge-summary) then `python3 bench/analysis/factorial_analysis.py`; runner `bench/factorial/run_factorial.py`; prereg-time live-index census `bench/factorial/live_decllexists_census.json` (3/100 golds resolvable in the live Brain index at prereg time — the drop-3 sensitivity row above).